Name ________________________ **Class: 14S**________ **Reg Number:** _____

MERIDIAN JUNIOR COLLEGE
JC 2 Preliminary Examination
Higher 2

Chemistry **9647/03**

Paper 3 Free Response **16 September 2015**

2 hours

Additional Materials: *Data Booklet*
Writing Paper

INSTRUCTIONS TO CANDIDATES

Write your name, class and register number in the spaces provided at the top of this page.

Answer 4 out of 5 questions in this paper.

Begin each question on a fresh page of writing paper.

Fasten the writing papers behind the given **Cover Page for Questions 1, 2 & 3** and **Cover Page for Questions 4 & 5** respectively.

Hand in Questions **1**, **2** & **3** and **4** & **5** **separately**.

You are advised to spend about 30 minutes per question only.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

Answer any 4 out of 5 questions in this paper.
*Begin each question on a **fresh sheet** of writing paper.*

1 In a blast furnace to produce iron from iron ore, large quantities of coke, C, and iron ores, Fe_2O_3, are added to the top of the furnace. Carbon dioxide is also produced in the process of the reaction.

The data for the enthalpy change of formation and entropy for the reactants and products in the blast furnace are shown below.

Species	ΔH_f^θ / kJ mol^{-1}	S^θ / J mol^{-1} K^{-1}
Coke, C(s)	0	5.69
Carbon dioxide, CO_2(g)	–393.51	213.68
Iron, Fe(s)	0	27.28
Iron ores, Fe_2O_3(s)	–824.25	87.40

(a) **(i)** Write a balanced equation including state symbols, for the production of pure iron in the blast furnace. [1]

(ii) Explain what is meant by *entropy*. [1]

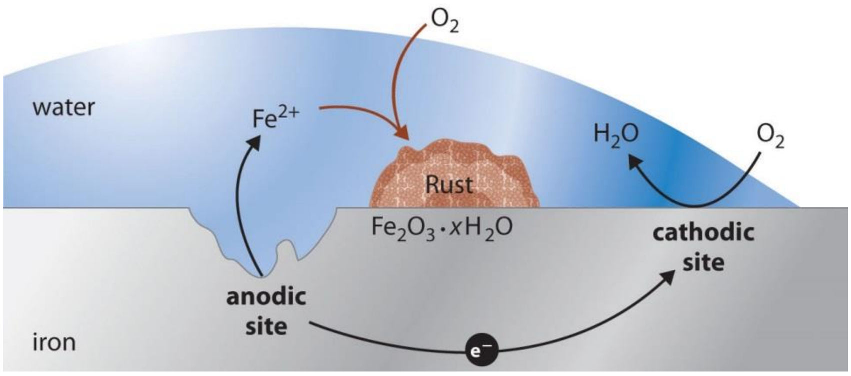
(iii) The entropy change for the production of **one mole** of pure iron is +140 J mol^{-1} K^{-1}. Explain the significance of the sign of the entropy change of the reaction. [1]

(iv) Using the data provided, calculate the enthalpy change, ΔH, to produce **one mole** of pure iron for the reaction in **(a)(i)**. [1]

(v) Explain the effect of low temperature on the spontaneity of the reaction and determine the minimum temperature for the reaction to occur. [2]

Rust, $Fe_2O_3 \cdot xH_2O$, is another form of iron(III) oxide. It is formed due to the corrosion of the iron metal when it comes in contact with water and oxygen.

During the reaction, iron is oxidised at an anodic site on the surface of the iron to form iron(II) ions while oxygen is reduced to water at the cathodic site. Electrons are transferred from the anode to the cathode through the electrically conductive metal. Rust, $Fe_2O_3 \cdot xH_2O$ is formed by the subsequent oxidation of iron(II) ions by atmospheric oxygen.



(b) **(i)** Write suitable half-equations and hence construct an ionic equation to show the oxidation of iron metal with oxygen. [2]

(ii) Using data from the *Data Booklet*, calculate the standard cell potential of the reaction in **(b)(i)**. [1]

(iii) Carbon dioxide readily dissolves in water to from carbonic acid, H_2CO_3.

$$CO_2 + H_2O \rightleftharpoons H_2CO_3$$

Using answers from **(b)(i)** and **(b)(ii)**, suggest why the presence of carbon dioxide in the atmosphere will cause the rusting of iron to be more feasible. [2]

(iv) It is possible to protect iron from rusting by attaching another metal onto it. Using data from *Data Booklet*, explain how zinc could protect iron from rusting. [2]

(v) Suggest an explanation why iron rusting is a slow process. [1]

(c) **(i)** It was found that the overall kinetics for rusting of iron is first order with respect to $[H^+]$. The rate of rusting is 3.16×10^{-5} mol dm^{-3} min^{-1} and $[H^+] = 0.4$ mol dm^{-3} at 25 ^oC.

Calculate the half-life of the reaction. [2]

(ii) Using an appropriate sketch of the Maxwell-Boltzmann distribution diagram, explain how an increase in temperature would affect the rate of rusting of iron. [4]

[Total: 20]

2 **(a)** The decomposition of calcium ethanoate takes place between 300 ^oC and 400 ^oC.

$$Ca(CH_3CO_2)_2 \rightarrow CaCO_3 + CH_3COCH_3$$

(i) Draw the dot-and-cross diagram for a carbonate ion. [1]

(ii) A 2.00 g impure sample of calcium ethanoate was strongly heated to about 400 ^oC until no further change in mass was observed. The mass of the only solid product remaining is 1.12 g.

Calculate the percentage purity of calcium ethanoate in the 2.00 g sample. [2]

(iii) Further decomposition takes place when calcium ethanoate is heated to about 900 ^oC. Suggest the decomposition products formed when calcium ethanoate is further heated to about 900 ^oC. [1]

(b) The decomposition of other calcium dicarboxylate salts may take place as follows:

$$Ca[O_2C(CH_2)_nCO_2] \rightarrow CaCO_3 + (CH_2)_nCO,$$

where $(CH_2)_nCO$ is a cyclic compound and $n \geq 2$

The thermal decomposition temperature of calcium dicarboxylate decreases as the alkyl chain increases (i.e. *n* increases).

(i) Suggest the structure of the organic compound formed when calcium hexanedioate, $Ca[O_2C(CH_2)_4CO_2]$, is decomposed. [1]

(ii) Explain why the thermal decomposition temperature of calcium dicarboxylates decreases as *n* increases. [2]

(c) The melting points and theoretical lattice energy values of oxides and carbonates of magnesium are as follows:

compound	**melting point / °C**	**theoretical lattice energy / kJ mol^{-1}**
$MgCO_3$	540	–3120
MgO	2850	–3800

(i) Define *lattice energy*. [1]

(ii) In terms of structure and bonding, explain the difference in melting points between magnesium carbonate and magnesium oxide. [2]

(iii) Using the following information and relevant data from the *Data Booklet*, draw the Born-Haber cycle and calculate the experimental lattice energy of magnesium oxide.

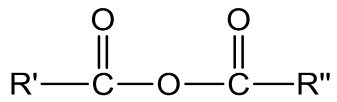
enthalpy change of formation of magnesium oxide	–642 kJ mol^{-1}
enthalpy change of atomisation of magnesium	+148 kJ mol^{-1}
sum of first and second electron affinity of oxygen	+602 kJ mol^{-1}

[4]

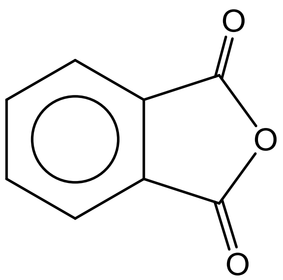
(iv) The theoretical lattice energy of magnesium oxide is –3800 kJ mol^{-1}, which is in close agreement with the experimental value calculated in **(c)(iii)**.

However, the theoretical and experimental lattice energy values of magnesium carbonate differ considerably. Suggest a reason for this difference. [1]

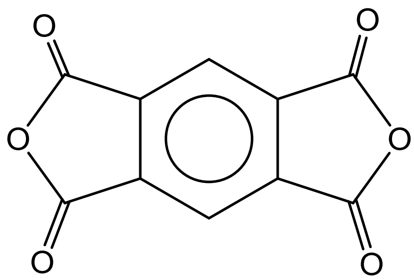
(d) Acid anhydrides are carboxylic acids derivatives with the following functional groups.



Phthalic anhydride and pyromellitic dianhydride have the following structure.



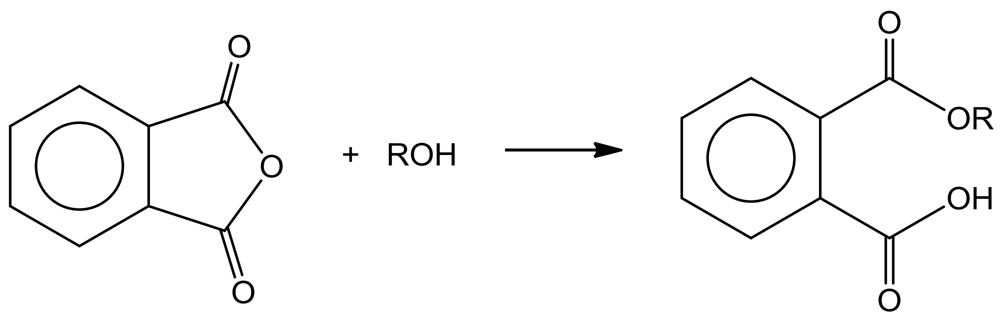
phthalic anhydride



pyromellitic dianhydride

(i) Suggest the structure of the organic product formed when phthalic anhydride is heated with aqueous hydrochloric acid. [1]

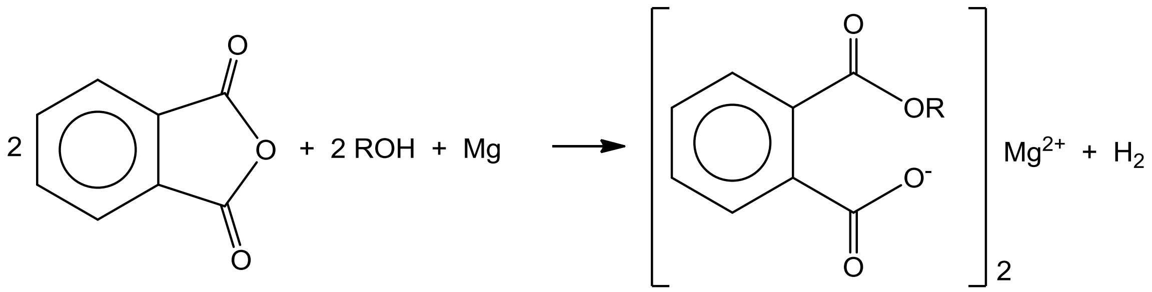
(ii) When phthalic anhydride is reacted with an alcohol, ROH, a compound with an ester group and a carboxylic acid group is produced.



1. State the type of reaction when phthalic anhydride is reacted with an alcohol. [1]

2. From your understanding of the reaction for phthalic anhydrides, draw the **two** isomers formed when one mole of pyromellitic dianhydride is reacted exactly with two moles of alcohol, ROH. [2]

(iii) In the presence of magnesium, the reaction of phthalic anhydride and alcohol in **(d)(ii)** was known to take place at a faster rate. The equation may be represented as shown:

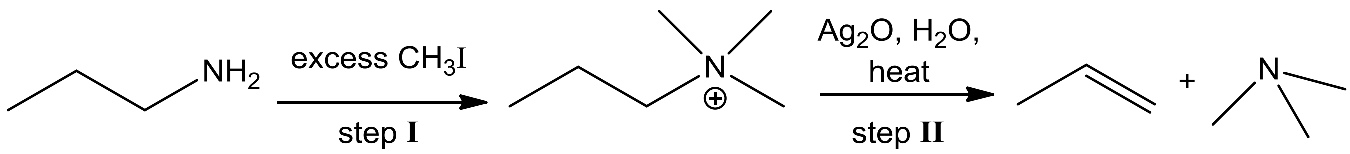


Suggest how magnesium could speed up the reaction of phthalic anhydride and alcohol. [1]

[Total: 20]

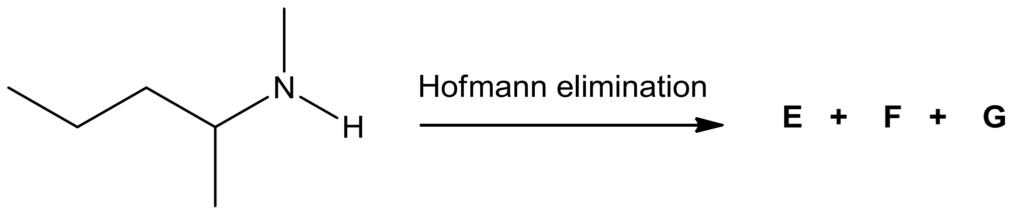
3 Alkenes have many applications in chemical industry. They are usually used as starting materials in the syntheses of alcohols, plastics, detergents, and fuels. Alkenes can be produced easily from cracking of hydrocarbons and elimination of organic compounds.

(a) Hofmann elimination is an organic reaction used to convert an amine to an alkene using methyl iodide, silver oxide and water under thermal conditions. An example is shown below for the 2-step conversion of propylamine to yield propene.



(i) Suggest the role of CH_3I in step **I**. [1]

(ii) When (1,N)-dimethylbutylamine undergoes the same 2-step reaction, three alkenes **E**, **F** and **G** are formed.



(1,N)-dimethylbutylamine

alkenes

Draw the structures of **E**, **F** and **G**. [3]

(iii) In another Hofmann elimination experiment, an alkene formed undergoes hydrogenation to form alkane **H**, C_6H_{14}. Compound **H** reacted with chlorine in the presence of ultraviolet light to form only **two** monochlorinated compounds.

During free-radical substitution of alkanes, different types of hydrogen atoms are replaced by chlorine atoms at different rates as shown in the following table.

Type of hydrogen atom	Reaction	Relative rate
primary	$RCH_3 \rightarrow RCH_2Cl$	1
secondary	$R_2CH_2 \rightarrow R_2CHCl$	7
tertiary	$R_3CH \rightarrow R_3CCl$	21

Using the above information and considering the number and type of hydrogen atoms within the molecule **H**, draw the structures of the **two** monochlorinated isomers formed and predict their relative ratio. [3]

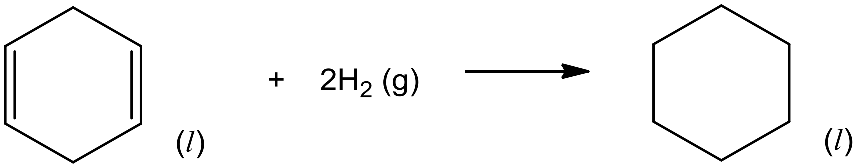
(b) Cyclohexane, C_6H_{12}, and cyclohexa-1,4-diene, C_6H_8, are commonly used as fuels. The standard enthalpy change of combustion of cyclohexane is –3920 kJ mol^{-1}.

(i) Calculate the minimum mass of cyclohexane required to increase the temperature of 100 g of water from 0 ^{o}C to 100 ^{o}C. Assume that the process is 80% efficient and that the specific heat capacity of water is 4.18 J g^{-1} K^{-1}. [2]

(ii) Explain why the mass of cyclohexane required to increase the temperature of 100 g of ice from 0 ^{o}C to 100 ^{o}C is likely to be higher than that calculated in **(b)(i)**, assuming the same process efficiency. [1]

(iii) The heat released when 1 g of a substance undergoes combustion is known as its fuel value in kJ g^{-1}. Calculate the fuel value of cyclohexane. [1]

(iv) Cyclohexa-1,4-diene, C_6H_8, can undergo catalytic hydrogenation to form cyclohexane, C_6H_{12}.



With the use of the following data, construct an energy cycle and use it to calculate the enthalpy change for the hydrogenation of liquid cyclohexa-1,4-diene to **one** decimal place.

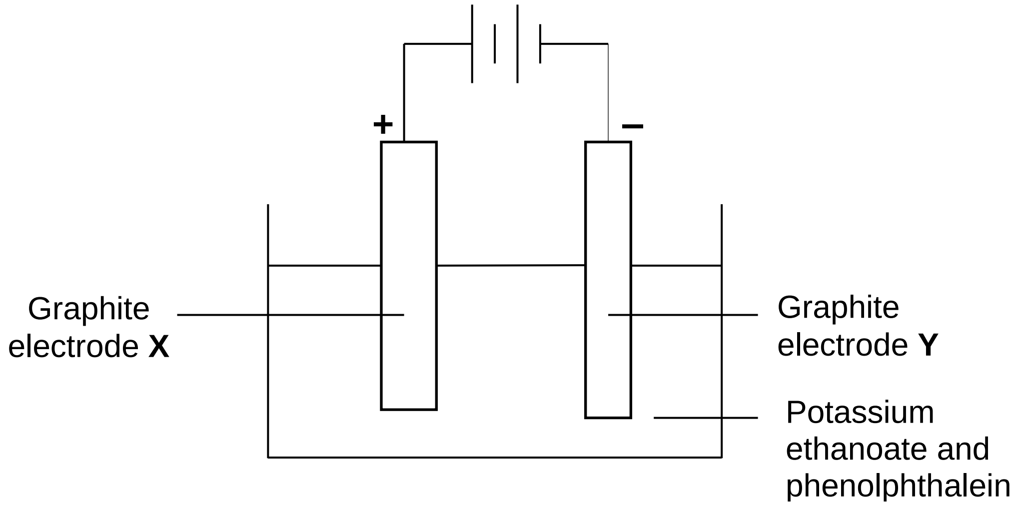
	ΔH / kJ mol^{-1}
Enthalpy change of combustion of liquid cyclohexane, ΔH_c (C_6H_{12})	–3920
Enthalpy change of combustion of gaseous cyclohexa-1,4-diene, ΔH_c (C_6H_8)	−3540
Enthalpy change of vapourisation of cyclohexa-1,4-diene, ΔH_{vap} (C_6H_8)	+34.3
Enthalpy change of combustion of hydrogen gas, ΔH_c (H_2)	−286

[3]

(c) Potassium ethanoate can be used to prepare ethane by an electrochemical reaction, which is known as the Kolbe electrolysis reaction. Ethane is formed at the anode by the following reaction:

$$2CH_3CO_2^- (aq) \rightarrow C_2H_6 (g) + 2CO_2 (g) + 2e$$

A chemist investigated the electrolysis of aqueous solution of potassium ethanoate containing phenolphthalein, using graphite electrodes in the apparatus as shown below.



(i) State and explain if any colour change is expected at electrode **Y**, and construct an equation for the **overall** reaction with state symbols. [2]

(ii) 3.60 dm^3 of ethane at room temperature and pressure was produced when a current was allowed to pass through a solution of potassium ethanoate for 15 minutes.

Calculate the current required for the above electrolysis process and the total volume of gases produced at electrodes **X** and **Y**. [2]

(iii) Real gases deviate from ideal gas behaviour by different extents. Deduce which gas, ethane or carbon dioxide, formed at the anode would behave more ideally. Explain your answer. [2]

[Total: 20]

4 Fish contains fatty acids which can undergo autoxidation to release aldehydes with exceptionally strong smells. Fish is also a major source of iron which is an important nutrient in the body.

(a) An example of such an aldehyde in the fatty acids of fish is pentanal, $C_5H_{10}O$, which has a pungent almond smell.

Describe the mechanism when pentanal is reacted with HCN under suitable conditions. Show all curly arrows, charges, dipoles and any relevant lone pairs. [3]

(b) **J** is an isomer of pentanal, $C_5H_{10}O$. **J** forms a colourless gas with sodium metal as well as a pale yellow precipitate with alkaline aqueous iodine. On heating **J** with acidified potassium manganate(VII), only one organic product **K**, C_5H_8O is formed which gives orange crystals with 2,4-dinitrophenylhydrazine.

Deduce, with reasoning, the structures for **J** and **K**. [4]

(c) *The use of the Data Booklet is relevant to this question.*

A chemist was given an unknown solution **A** containing an aqueous solution of iron ions. A series of experiments were conducted to determine the oxidation state of iron in solution **A**.

In Experiment **1**, aqueous sodium hydroxide was added to an equal volume of solution **A**. A green precipitate was observed. On standing, the green precipitate turned red–brown.

In Experiment **2**, when chlorine gas was bubbled into solution **A**, a yellow solution **B** was obtained. On reacting solution **B** with an aqueous solution of potassium thiocyanate, KSCN, a blood red solution **C,** $[Fe(SCN)(H_2O)_5]^{2+}$, was obtained.

(i) State the oxidation number of iron in solution **A**. Explain the observations of Experiment **1** as fully as you can, and support your answers with relevant equations. [3]

(ii) Using relevant E^θ values from the *Data Booklet*, explain the formation of solution **B** in Experiment **2**. Construct a balanced equation for this reaction. [2]

(iii) What type of reaction occurs when the blood red solution **C** is formed from yellow solution **B**? [1]

(iv) Draw the structure of $[Fe(SCN)(H_2O)_5]^{2+}$ present in solution **C**, showing clearly the shape of the ion. [1]

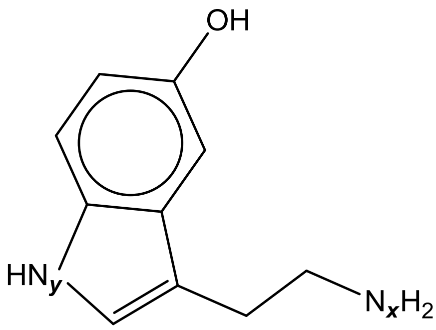
(v) Suggest an explanation for the red coloration of $[Fe(SCN)(H_2O)_5]^{2+}$. [3]

(vi) Explain, with the aid of equations and reference to relevant E^{θ} values, how solution **A** can catalyse the reaction between iodide and peroxodisulfate ions, $S_2O_8^{2-}$. [3]

[Total: 20]

5 Alkaloids are a group of naturally occurring chemical compounds that contain mostly basic nitrogen atoms. They have a wide range of pharmacological activities including anticancer and antibacterial uses.

(a) *Serotonin*, $C_{10}H_{12}N_2O$, a neurotransmitter, is an example of an alkaloid. It is primarily found in the gastrointestinal tract and the central nervous system of humans.



Serotonin

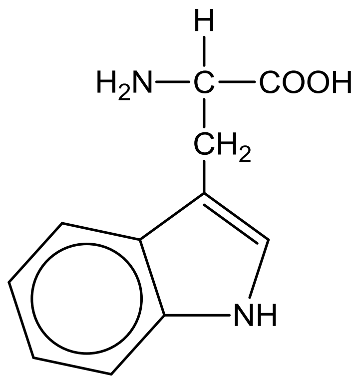
(i) Explain why the pK_b value for N_x is lower than N_y atom in *Serotonin*. [2]

(ii) Draw the structures of the organic products formed when *Serotonin* is reacted with the following under suitable conditions:

1. ethanoyl chloride [2]

2. aqueous bromine [1]

(b) *Serotonin* is synthesized from the amino acid *L-tryptophan*, one of the common natural amino acids, by a short metabolic pathway.



L-tryptophan

There are **only** two pK_a values associated with *L-tryptophan*: 2.38 for the α-carboyxl group and 9.39 for the α–amino group.

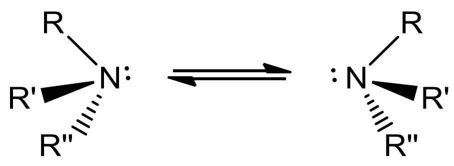
In an experiment, 10.0 cm^3 of 0.100 mol dm^{-3} of the protonated form of *L-tryptophan* solution is titrated with 30 cm^3 of 0.100 mol dm^{-3} NaOH(aq).

(i) Calculate the initial pH of 10.0 cm^3 of 0.100 mol dm^{-3} of *L-tryptophan*. [1]

(ii) Hence, sketch the pH-volume added curve you would expect to obtain from the above titration.

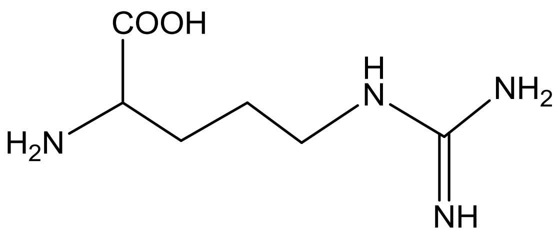
Briefly describe how you have calculated the key points on the curve. [4]

(c) In chemistry, a nitrogen compound with trigonal pyramidal shape such as the following amine can undergo rapid nitrogen inversion.



The nitrogen atom in this case exhibits chirality. This is because the lone pair of electrons on the nitrogen effectively behaves as a bonding pair of electrons of an atom.

Similarly, nitrogen atom may also exhibit cis-trans isomerism as shown in *Arginine*.



Arginine

(i) Using the context given above, draw the cis-trans isomers involving the N atom of *Arginine*. [2]

(ii) Calculate the total number of stereoisomers exhibited by *Arginine*. [1]

(d) Compound **P**, $C_{10}H_{12}N_2O$, an isomer of *Serotonin*, is found to undergo the following reactions.

P is insoluble in water and NaOH(aq) but it dissolves readily in HC*l*(aq).

On heating **P** with NaOH(aq), a pungent gas is evolved which turns moist red litmus paper blue.

P reacts with aqueous bromine to form **Q**, $C_{10}H_{10}N_2O_2Br_4$.

P undergoes oxidation with hot acidified $KMnO_4$ to form CO_2 gas and **R**, $C_8H_7NO_4$.

R is formed by heating 5-nitro-1,3-benzenedicarboxylic acid with tin and concentrated hydrochloric acid.

Suggest the structures for compounds **P**, **Q**, and **R** and explain your reasoning fully. [7]

[Total: 20]

END OF PAPER